Notes: Atkeson and Kehoe (JPE, 2005)
Modeling and Measuring Organization Capital

Contents

1 One-sentence summary

Atkeson and Kehoe build and calibrate a plant life-cycle model in which young plants accumulate organization-specific knowledge over time, and they use U.S. manufacturing data to infer that payments to organization capital are quantitatively large–about 3.3 percent of manufacturing output and roughly 37 percent of net payments to physical capital in their benchmark calibration.

2 Big picture

2.1 Question

How large are the payments to organization capital in U.S. manufacturing, and can those payments be inferred from the observed life cycle of manufacturing plants?

2.2 Core idea

Manufacturing plants are born small, grow with age, and eventually die. The paper interprets this life cycle as the outcome of accumulating plant-specific organizational knowledge. Owners of plants pay start-up and operating costs and wait for plants to grow. The payments they later receive are compensation for this intangible stock of organization capital.

2.3 Why plant age matters

Plant age is informative because the model links age to accumulated knowledge. Young plants have frontier technology but little built-up knowledge. Older surviving plants have accumulated organization capital, so they generate higher organization rents.

2.4 Main quantitative result

The benchmark model implies that payments to owners of organization capital equal about 3.3 percent of manufacturing output. This is about 41 percent of the implied payments to all intangible capital in U.S. manufacturing and about 37 percent of net payments to physical capital.

2.5 Economic takeaway

A substantial part of measured output should be understood as compensation for intangible plant-level organization capital rather than only for labor and physical capital. This changes how we interpret profits, intangible capital, and the value of mature productive organizations.

3 Incremental contribution of the paper

3.1 A quantitative measure of organization capital

Earlier work discussed plant-specific knowledge qualitatively. This paper converts that idea into a quantitative growth model that produces a measurable flow of payments to organization capital.

3.2 Plant life-cycle facts as measurement discipline

The paper uses plant age profiles of employment, job creation, job destruction, and survival to discipline the accumulation process. The measurement of intangible capital is therefore anchored in micro plant dynamics rather than only aggregate residuals.

3.3 Organization capital as claims to future rents

The paper clarifies the economics of organization capital: owners of young plants accept low early rents because growth generates future rents. Organization rents are the return on the owners' waiting cost and start-up investment.

3.4 Bridge between industry dynamics and growth accounting

The paper connects industry evolution models with macroeconomic income accounting. It explains how a plant-level life-cycle model can be used to decompose aggregate manufacturing output into payments to workers, managers, physical capital, and organization capital.

3.5 Benchmark for intangible-capital debates

By comparing its organization-capital estimate to McGrattan and Prescott's broader intangible-capital measure, the paper gives a useful benchmark: organization capital is only one component of intangibles, but it can account for a large share of intangible payments.

4 Data, calibration, and measurement

4.1 Plant-level life-cycle moments

The calibration is disciplined by U.S. manufacturing plant facts, especially from Davis, Haltiwanger, and Schuh. The key moments are:

- employment shares by plant age;
- job creation rates by age;
- job destruction rates by age;
- job destruction among failing plants by age;
- distribution of job creation and destruction by growth-rate bins;
- average productivity by age.

Interpretation: These moments identify how fast plants grow, how often they exit, how much turnover comes from young versus old plants, and whether productivity levels are systematically higher for older plants.

4.2 What counts as organization capital

Organization capital is the pair consisting of plant-specific productivity and plant age. A young plant begins with frontier technology but little knowledge about how to operate it. Over time, stochastic learning raises plant-specific productivity.

4.3 Aggregate output shares

The paper divides output payments into four groups: workers, managers, physical capital, and organization capital. The model is calibrated so that the physical capital share matches U.S. manufacturing data. The residual payments implied by the model include payments to managers and owners of organization capital.

4.4 Benchmark against broader intangible capital

The paper compares its organization-capital payment estimate with an aggregate measure of payments to all intangible capital. In manufacturing, payments to intangible capital are about 8 percent of output, and the model's organization-capital payments are about 40 percent of that amount.

5 Model setup and derivations

5.1 Production and factors

Each plant operates a production technology using physical capital and labor. Plant-specific productivity evolves over age. The paper uses a Cobb-Douglas environment in which the return to physical capital can be calibrated to match aggregate capital payments.

5.2 Plant productivity dynamics

A plant's productivity grows stochastically with age. The mean and variance of productivity shocks vary with plant age. The estimated process is chosen to match the plant life-cycle moments in the data.

5.3 Start-up and operating costs

Plant owners pay start-up costs to create plants and fixed costs to operate them. A new plant is valuable only if expected future rents cover the cost of entry. Free entry pins down the value of a new plant.

5.4 Organization rents

Organization rents are variable profits net of operating costs and after paying physical capital and labor. Young plants often have low rents; older successful plants earn higher rents because they have accumulated organization capital.

5.5 Value of a plant

The value of a plant is the discounted value of expected future organization rents. A mature plant is valuable because it represents the right to receive rents generated by accumulated know-how.

5.6 Managerial span of control

The parameter ν governs the span of control. The benchmark is $\nu = 0.85$. Varying ν changes the split between manager payments and owner payments to organization capital, but the benchmark implication remains that organization capital is quantitatively important.

6 Calibration logic and model fit

6.1 Matching the plant age distribution

The model targets the strong empirical fact that most manufacturing employment is in old plants, while young plants account for a disproportionate share of entry and growth. Figure 1 displays this fit.

6.2 Matching job creation and destruction

A good model must generate both fast growth among surviving young plants and high failure rates among new plants. The calibration uses age-specific job creation and destruction to discipline this tradeoff.

6.3 Matching growth-rate distributions

The paper also evaluates the model against the cross-sectional distribution of job creation and destruction by growth-rate bins. This tests whether the model reproduces the tails of plant-level growth, not only age averages.

6.4 Average productivity by age

The model's key assumption does not imply that older plants should have sharply higher measured average productivity. Figure 5 shows that average productivity is fairly flat by age in the U.S. data, a pattern the model can accommodate.

7 Measurement of organization capital

7.1 Output decomposition

In the benchmark, the paper calibrates the physical capital share to match the data. The remaining output is divided between workers, managers, and owners of organization capital.

7.2 Benchmark estimate

For $\nu = 0.85$, the model implies:

- labor share around 76.8 percent;
- physical capital share around 19.9 percent;
- organization capital share around 3.3 percent;
- net payments to physical capital around 8.9 percent.

Interpretation: The organization-capital share is smaller than labor and physical capital shares, but it is large relative to net physical capital payments and to aggregate intangible payments.

7.3 Sensitivity to span of control

Changing ν from 0.80 to 0.90 changes the organization-capital share from about 4.4 percent to 2.2 percent. The benchmark conclusion is robust: organization-capital rents are economically meaningful across plausible values.

8 Identification-style interpretation

8.1 What identifies organization capital in the model

Organization capital is not directly observed. It is inferred from the combination of:

1. plant life-cycle growth and exit patterns;
2. start-up and operating cost structure;
3. free-entry valuation of new plants;
4. aggregate output shares;
5. the model-implied value of waiting for plant productivity to grow.

8.2 Why life-cycle facts matter

If plants did not grow systematically with age, the organization-capital interpretation would be weak. The observed life cycle is the empirical object that makes the organization-capital stock measurable.

8.3 Main threats and limitations

The measurement depends on the model's assumptions about plant-level productivity, fixed costs, stochastic growth, and span of control. It does not directly observe plant-specific knowledge, managerial practices, or organizational routines. The result should therefore be read as a model-implied accounting measure rather than a direct empirical observation.

8.4 Why the model is still informative

The model is useful because it disciplines an otherwise unmeasured intangible asset using observable plant dynamics. It converts a broad idea–firms have accumulated know-how–into a quantitative object linked to aggregate income shares.

9 Tables and figures

9.1 Figure 1: Employment statistics by manufacturing plant age in the model and in the 1988 U.S. data

Read: Figure 1 compares model and data for employment, job creation, and job destruction by plant age. It shows that the model matches the concentration of employment in old plants and broadly matches age patterns of job creation and destruction.

Economic message: The figure validates the core life-cycle mechanism. If the model could not reproduce plant age patterns, its measurement of organization capital would not be credible.

Detailed interpretation: The most important panel is employment by age: more than half of employment is in plants older than 25 years. This means that mature organizations account for a large share of productive capacity. Job creation and destruction panels show that young plants are volatile while mature plants dominate the stock of employment.

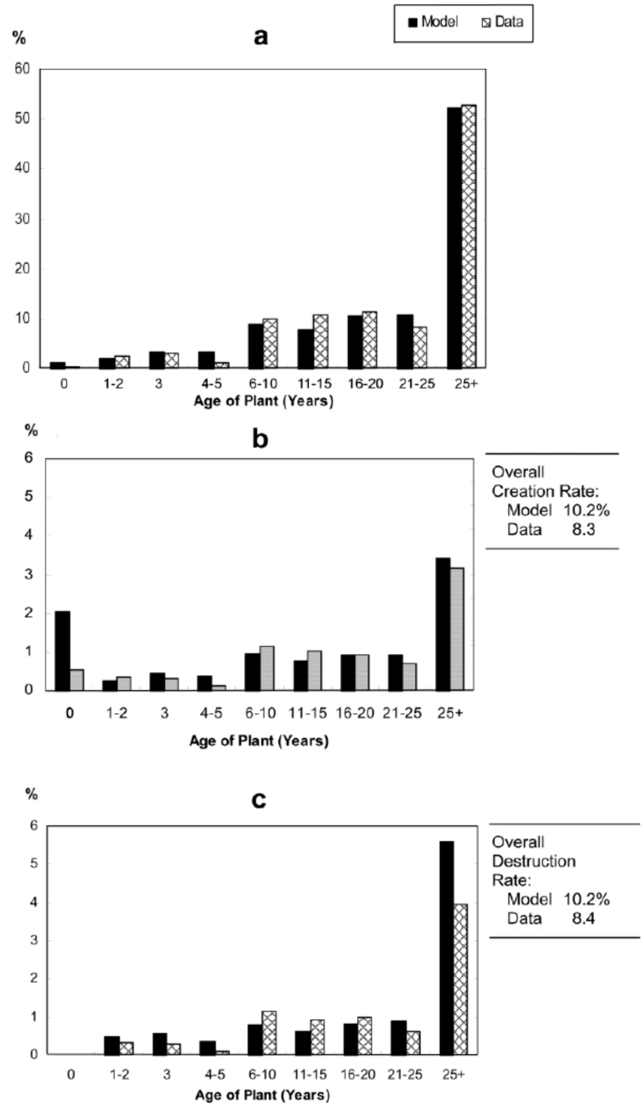


FIG. 1.—Employment statistics by manufacturing plant age in the model and in the

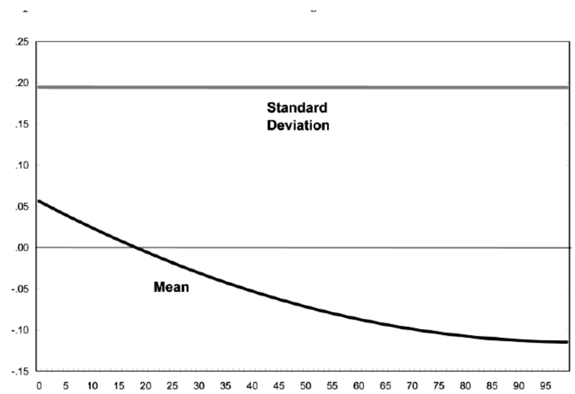
Figure 1: Employment statistics by manufacturing plant age in the model and in the 1988 U.S. data. Single unpaired visual, set to width=0.6\linewidth as requested. The three panels compare model and data for employment, job creation, and job destruction by plant age.

9.2 Figure 2: Means and standard deviations of shocks to plant size by age of plant; Figure 3: Job destruction in failing plants by age of plant

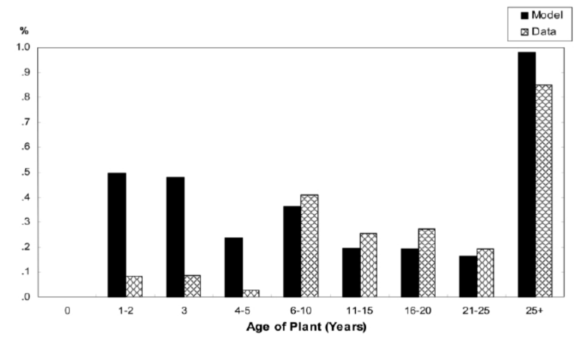
Read: Figure 2 shows the calibrated age profile of shocks to plant size. The mean falls with age, while the standard deviation is roughly flat. Figure 3 compares model and data for job destruction in failing plants by age.

Economic message: These two figures jointly discipline the stochastic learning and exit process. Young plants need high growth potential but also high failure risk, while older plants are larger and more stable.

Detailed interpretation: Figure 2 is the mechanism behind the plant life cycle: young plants have positive growth potential; older plants have less growth. Figure 3 checks whether failure events generate the correct contribution to job destruction across age groups.



(a) Figure 2: shocks to plant size by age.



(b) Figure 3: job destruction in failing plants by age.

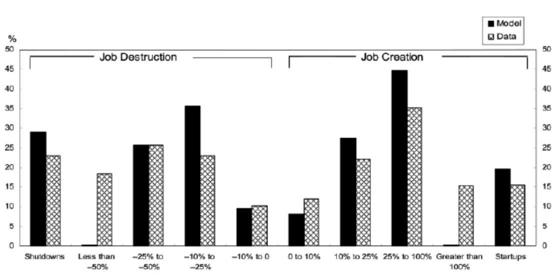
Figures 2 and 3. Adjacent related figures are placed on the same row. Figure 2 shows the calibrated shock process; Figure 3 tests the model against failure-related job destruction.

9.3 Figure 4: Distribution of job creation and destruction; Figure 5: Average productivity of plants by age

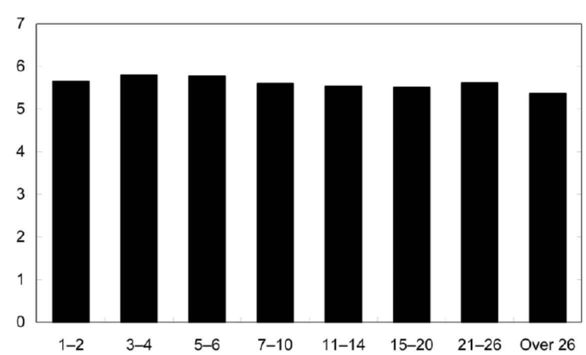
Read: Figure 4 compares model and data for the distribution of job creation and destruction by growth-rate categories. Figure 5 shows that average plant productivity is fairly flat across plant age groups.

Economic message: Figure 4 checks whether the model matches growth-rate dispersion, while Figure 5 addresses a possible concern: organization capital does not require older plants to have much higher measured average productivity.

Detailed interpretation: Figure 4 shows that the model captures key parts of the job-flow distribution but also has differences in tails. Figure 5 is important conceptually: organization capital is not simply measured productivity. It is the value of the accumulated operating knowledge and future rents embodied in the plant life cycle.



(a) Figure 4: distribution of job creation and destruction.



(b) Figure 5: average productivity by plant age.

Figures 4 and 5. Adjacent related figures are placed on the same row. They evaluate model fit for job-flow distributions and measured productivity by age.

9.4 Table 1: Accounting for Output in the U.S. Manufacturing Sector

Read: Table 1 decomposes manufacturing output into labor, worker, manager, physical capital, intangible capital, organization capital, and net payments to physical capital. It reports both U.S. manufacturing data and model values for different span-of-control parameters ν.

Economic message: The table is the paper's main measurement result. In the benchmark $\nu = 0.85$, organization capital receives about 3.3 percent of output, which is large relative to net physical-capital payments.

Detailed interpretation: Table 1 connects the micro life-cycle model to aggregate accounting. The key comparison is not organization capital versus gross physical capital payments; it is organization capital versus the net payments to physical capital and the broader intangible-capital residual. This is where the model's estimate becomes economically large.

TABLE 1
ACCOUNTING FOR OUTPUT IN THE U.S. MANUFACTURING SECTOR

	Data on U.S. Manufacturing*	Model		
		$\nu = .80$	$\nu = .85$	$\nu = .90$
	Shares of Output (%)			
Labor	72.2	75.7	76.8	77.9
Workers	. . .	60.1	65.1	70.1
Managers	. . .	15.6	11.7	7.8
Physical capital	19.9	19.9	19.9	19.9
Intangible capital	8.0	. . .	. . .	. . .
Organization capital	. . .	4.4	3.3	2.2
	Other Payments			
Net payments to physical capital†	7.3	8.9	8.9	8.9

* U.S. manufacturing data are described in App. A.
† Net payments to physical capital are measured net of new investment, i.e., $rk - x$.

Table 1: Accounting for Output in the U.S. Manufacturing Sector. Single unpaired table, set to width=0.6\linewidth as requested. The table reports the aggregate output-share decomposition in the data and in the calibrated model.

10 Short summary

- Manufacturing plants have a pronounced life cycle: entry, growth, maturity, and exit.
- The paper interprets this life cycle as evidence of accumulated organization capital.
- Organization capital is plant-specific knowledge embodied in age and productivity.
- A calibrated industry-dynamics model maps plant life-cycle facts into aggregate payments to organization capital.
- The benchmark estimate is about 3.3 percent of manufacturing output.
- This is about 40 percent of broader intangible-capital payments in manufacturing.
- The estimate is also about 37 percent of net payments to physical capital.
- The model matches employment by age and key job-flow moments reasonably well.
- The paper's contribution is to make organization capital measurable using plant dynamics rather than treating it as an unobserved residual.

11 Conclusion

11.1 Core conclusion

Atkeson and Kehoe conclude that organization capital is quantitatively important in U.S. manufacturing. The life cycle of plants implies that owners of mature plants receive rents generated by accumulated organization-specific knowledge. These rents are large enough to matter for aggregate accounting.

11.2 Economic intuition

Young plants start small and grow as they learn how to use their technology. Owners pay entry and fixed operating costs and wait for plants to mature. The compensation for this waiting and organizational accumulation appears as organization rents.

11.3 Why the paper matters

The paper provides a bridge between micro plant dynamics and macro intangible capital measurement. It shows how plant age profiles can be used to measure the value of intangible organizational knowledge that is not directly recorded in national accounts.

11.4 Incremental contribution revisited

The paper's distinctive contribution is to treat plant life-cycle dynamics as a measurement device for organization capital. Rather than imputing all intangible capital from aggregate residuals, it uses a calibrated industry-evolution model to infer one specific intangible component.

11.5 Limitations

The estimate depends on the assumed productivity process, fixed costs, free entry, and the span-of-control parameter. The paper does not directly observe organization capital, so the result is model-implied. It also focuses on manufacturing, where plant-level dynamics are more directly measured than in services.

11.6 Takeaway

Organization capital is not just a metaphor for good management or firm culture. In this paper, it is a measurable asset whose rents can be inferred from how plants enter, grow, survive, and die.